

Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta function. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$; for any constant $c > c_*$, define the shifted phase $\Theta(t) := \theta(t) + ct$. Then $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection. The unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}, \quad (1)$$

with the principal (positive) square root of the strictly positive real quantity $\Theta'(\Theta^{-1}(u))$, is a real-valued function on $\mathbb{R}$ whose Fourier support is contained in $[-1, 1]$. By the Paley–Wiener theorem, Y extends to an entire function of exponential type ≤ 1 . Non-negativity of the spectral measure $S := |\widehat{Y}|^2$ together with Akhiezer’s theorem places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. On $\mathbb{R}$ the identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is the rearrangement of (1), with $\sqrt{\Theta'(t)} > 0$. This unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$ is a bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$ with preservation of multiplicity. The critical-line identity $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ for $t \in \mathbb{R}$, combined with reality of zeros of Y and the unitary transfer, fixes the zeros of ζ on the critical line as the Θ^{-1} -image of the real zeros of Y .

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-valued on $\mathbb{R}$. The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$ ties the zeros of Z on $\mathbb{R}$ to the zeros of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$ under $s = \frac{1}{2} + it$: the map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$, and $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$ makes the two sides vanish simultaneously.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing on $[0, \infty)$ but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ' . Monotonicity on all of $\mathbb{R}$ is restored by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, which enforces $\Theta' > 0$ on $\mathbb{R}$ and makes $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a C^∞ bijection. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds β_n , are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y : $\operatorname{supp} S \subseteq [-1, 1]$. Paley–Wiener then gives Y entire of exponential type ≤ 1 ; Akhiezer gives reality of its complex zeros. The real-line identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is a unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$, and so is a zero-set bijection with multiplicity preservation between Y on $\mathbb{R}$ and Z

on $\mathbb{R}$ (Theorem 3). The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ on $t \in \mathbb{R}$ yields the conclusion for the zeros of ζ on the critical line.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

For $t \in \mathbb{R}$, $\frac{1}{4} + \frac{it}{2}$ avoids $\mathbb{Z}_{\leq 0}$ (its real part is $\frac{1}{4}$), so $\theta, \theta', \theta''$ are C^∞ on $\mathbb{R}$.

Lemma 1 (Positivity of θ''). *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. $\square$

Lemma 2 (Exact value of $\theta'(0)$).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right) \quad (6)$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). $\square$

Definition 1 (Admissible constant).

$$c_* := -\theta'(0) = \frac{1}{2} \left(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi \right). \quad (7)$$

A real c is admissible if $c > c_*$. Throughout the paper c is admissible.

Lemma 3 (Monotone bijection). *For every admissible c , $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing C^∞ bijection.*

Proof. θ' is even on $\mathbb{R}$ (from (3); $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2})$ is even in τ) and by Lemma 1 $\theta'' > 0$ on $(0, \infty)$, so θ' is strictly increasing on $[0, \infty)$ and attains its minimum on $\mathbb{R}$ at $\tau = 0$, namely $\theta'(0) = -c_*$. For any $c > c_*$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ on $\mathbb{R}$. The standard asymptotic $\theta(\tau) = \frac{\tau}{2} \log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$ as $\tau \rightarrow \infty$ gives $\Theta(\tau) \rightarrow +\infty$ as $\tau \rightarrow +\infty$ and $\Theta(\tau) \rightarrow -\infty$ as $\tau \rightarrow -\infty$ (the second by the functional relation $\theta(-\tau) = -\theta(\tau)$ on $\mathbb{R}$). Strict monotonicity together with these limits gives that $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. Smoothness is inherited from $\theta \in C^\infty(\mathbb{R})$. $\square$

3 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u \in \mathbb{R}$, the unitary inverse phase pullback of Z is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (8)$$

The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ from Lemma 3; $\Theta'(\Theta^{-1}(u))$ is a strictly positive real number, and $\sqrt{\Theta'(\Theta^{-1}(u))}$ denotes its principal (positive) square root. The half-Jacobian factor renders the map $Z \mapsto Y$ an L^2 -isometry under the change of variable $u = \Theta(t)$.

Y is real-valued on $\mathbb{R}$ because Z is real-valued on $\mathbb{R}$ and the denominator is a strictly positive real number.

4 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n := \Theta'(2\pi n^2) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) - \frac{1}{2} \log \pi + c. \quad (9)$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 4 (Asymptotic of β_n from the exact form). *As $n \rightarrow \infty$,*

$$\beta_n = \log n + c + r_n, \quad |r_n| \leq \frac{1}{64} n^{-4} \text{ for all } n \geq 1. \quad (10)$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi, \quad (11)$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s|^2 = \frac{1}{16} + \pi^2 n^4$ yields

$$\operatorname{Re} \psi(s) = \frac{1}{2} \log\left(\frac{1}{16} + \pi^2 n^4\right) + r'_n = 2 \log n + \log \pi + r''_n, \quad (12)$$

where $|r''_n| \leq (B_2/2)|s|^{-2} + \dots \leq \frac{1}{32} n^{-4}$ for $n \geq 1$ by term-by-term bounding of the Bernoulli series. Therefore $\frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + i\pi n^2\right) = \log n + \frac{1}{2} \log \pi + r_n$ with $|r_n| \leq \frac{1}{64} n^{-4}$, and $\beta_n = \log n + c + r_n$. $\square$

Definition 4 (Truncated transform). *For $T > 0$ and $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du. \quad (13)$$

Lemma 5 (Exact Riemann–Siegel identity). *For every $t > 0$,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + (-1)^{N(t)-1} \left(\frac{t}{2\pi}\right)^{-1/4} R\left(\frac{t}{2\pi}\right), \quad (14)$$

with $N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor$, and with the remainder function $R(\tau)$ given exactly by the Siegel contour integral

$$R(\tau) = \frac{e^{-i\pi/8} e^{i\pi\tau}}{(2\pi)^{1/2} \tau^{1/4}} \cdot \frac{1}{2\pi i} \int_{0 \swarrow 1} \frac{e^{i\pi w^2/2 - 2\pi\sqrt{\tau} w}}{e^{i\pi w} - e^{-i\pi w}} dw, \quad (15)$$

in which the contour $0 \swarrow 1$ is the straight line of direction $e^{-i\pi/4}$ crossing the real axis between 0 and 1.

Proof. [4], §7.4, Theorem 7.1; [10]. □

Lemma 6 (Berry–Gabcke expansion of $R(\tau)$). *Set $p(\tau) := \sqrt{\tau} - N(2\pi\tau) - \frac{1}{2}$ for the centered fractional part, and*

$$\Psi(p) := \frac{\cos 2\pi(p^2 - p - \frac{1}{16})}{\cos 2\pi p}, \quad p \in (-\frac{1}{2}, \frac{1}{2}). \quad (16)$$

Then Ψ extends to a 1-periodic real-analytic function on $\mathbb{R}$ with absolutely convergent Fourier series

$$\Psi(p) = \sum_{m=0}^{\infty} a_m \cos(2\pi(m + \frac{1}{2})p), \quad (17)$$

whose Fourier coefficients satisfy $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$ and $\sum_{m \geq 0} |a_m| \leq 2.82$; and for every $\tau \geq 200/(2\pi)$,

$$R(\tau) = \Psi(p(\tau)) + E_0(\tau), \quad |E_0(\tau)| \leq 0.127 \tau^{-3/4}. \quad (18)$$

Proof. The closed form for $\Psi(p)$ and the Fourier-coefficient decay $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$ are [3], equations (16)–(18); the error bound $|E_0(\tau)| \leq 0.127 \tau^{-3/4}$ is [5], Theorem 4.2. Summing the rapidly decreasing series, $\sum_{m \geq 0} |a_m| \leq 4 \sum_{m \geq 0} e^{-\pi(m+1/2)^2/2} \leq 2.82$. □

Theorem 1 (Band-limitedness of Y). *For every $c > c_*$ and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$\lim_{T \rightarrow \infty} K_T(\mu) = 0. \quad (19)$$

Consequently the tempered Fourier transform $\hat{Y}$ is supported in $[-1, 1]$, and $\text{supp } S \subseteq [-1, 1]$, where $S := |\hat{Y}|^2$.

The theorem is assembled from two propositions: Proposition 1 controls each Dirichlet mode of the Riemann–Siegel main sum, and Proposition 2 controls the Riemann–Siegel remainder term.

Mode decomposition

Substitute (14) into (13) and expand $2 \cos \alpha = e^{i\alpha} + e^{-i\alpha}$:

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{+, -\}} \sum_{n=1}^{\lfloor \sqrt{T/(2\pi)} \rfloor} n^{-1/2} J_{n,\sigma}(T, \mu) + K_T^R(\mu), \quad (20)$$

where, setting $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T, \mu) := \int_{t_n}^T \sqrt{\Theta'(t)} e^{i\Phi_{n,\sigma,\mu}(t)} dt, \quad \Phi_{n,\sigma,\mu}(t) := \sigma \theta(t) - \sigma t \log n - \mu \Theta(t), \quad (21)$$

and

$$K_T^R(\mu) := \frac{1}{2\pi} \int_0^T (-1)^{N(t)-1} \left(\frac{t}{2\pi}\right)^{-1/4} R\left(\frac{t}{2\pi}\right) \sqrt{\Theta'(t)} e^{-i\mu \Theta(t)} dt. \quad (22)$$

Instantaneous frequency in the u -variable

Along $u = \Theta(t)$, with $du = \Theta'(t) dt$,

$$\frac{d}{du} \Phi_{n,\sigma,\mu}(\Theta^{-1}(u)) = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} - \mu. \quad (23)$$

For every $n \geq 1$, $\sigma \in \{+, -\}$ and $t \geq t_n$,

$$\left| \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} \right| < 1. \quad (24)$$

Proof of (24). The denominator equals $\Theta'(t) > 0$. The numerator's modulus satisfies $|\theta'(t) - \log n| < \theta'(t) + c$ iff $\log n - \theta'(t) < \theta'(t) + c$ and $\theta'(t) - \log n < \theta'(t) + c$. The second is $-\log n < c$, which holds for $n \geq 1$ and $c > c_* > 0$. The first is $\log n - c < 2\theta'(t)$. At $t = t_n$, $\theta'(t_n) = \beta_n - c = \log n + r_n$ with $|r_n| \leq \frac{1}{64}n^{-4}$ (Lemma 4), so $2\theta'(t_n) = 2\log n + 2r_n \geq 2\log n - \frac{1}{32}n^{-4} > \log n - c$ for $c > c_* > 0$ and $n \geq 1$. For $t > t_n$, $\theta'(t) > \theta'(t_n)$ (since $\theta'' > 0$ on (t_*, ∞) with $t_* < t_1$), so the inequality persists. $\square$

Consequently, for every μ with $|\mu| > 1$,

$$\left| \frac{d}{du} \Phi_{n,\sigma,\mu}(\Theta^{-1}(u)) \right| \geq |\mu| - 1 =: \lambda > 0, \quad (25)$$

uniformly in $n \geq 1$, $\sigma \in \{+, -\}$ and $u \geq \Theta(t_n)$.

Per-mode decay and tail assembly

Proposition 1 (Per-mode decay of the Dirichlet integrals). *Fix $c > c_*$ and $\mu \in \mathbb{R}$ with $|\mu| > 1$, and set $\lambda := |\mu| - 1$. There exist constants $C_1(\mu, c)$ and $C_2(\mu, c)$, independent of n , σ and T , such that for every $n \geq 1$, $\sigma \in \{+, -\}$ and $T \geq t_n$,*

$$|J_{n,\sigma}(T, \mu)| \leq \frac{C_1(\mu, c)}{\Theta'(T)^{3/2}} + \frac{C_2(\mu, c)(1 + \log n)}{\beta_n^{3/2}}. \quad (26)$$

Moreover $J_{n,\sigma}(\infty, \mu) := \lim_{T \rightarrow \infty} J_{n,\sigma}(T, \mu)$ exists.

Proof. Pull back to $u = \Theta(t)$. Set $G_n(u) := 1/\sqrt{\Theta'(\Theta^{-1}(u))}$ and $\Psi_{n,\sigma,\mu}(u) := \Phi_{n,\sigma,\mu}(\Theta^{-1}(u))$:

$$J_{n,\sigma}(T, \mu) = \int_{\Theta(t_n)}^{\Theta(T)} G_n(u) e^{i\Psi_{n,\sigma,\mu}(u)} du. \quad (27)$$

By (25), $|\Psi'_{n,\sigma,\mu}(u)| \geq \lambda > 0$ on $[\Theta(t_n), \infty)$. Two integrations by parts give

$$J_{n,\sigma}(T, \mu) = B_0(T) - B_0(t_n) + B_1(T) - B_1(t_n) + \int_{\Theta(t_n)}^{\Theta(T)} \frac{d}{du} \left[\frac{1}{i\Psi'} \frac{d}{du} \left(\frac{G_n}{i\Psi'} \right) \right] e^{i\Psi} du, \quad (28)$$

where $B_0(s) := G_n(\Theta(s))e^{i\Psi(\Theta(s))}/(i\Psi'(\Theta(s)))$, $B_1(s) := -(i\Psi')^{-1}(d/du)(G_n/(i\Psi'))|_{u=\Theta(s)} e^{i\Psi(\Theta(s))}$.

Using $t = \Theta^{-1}(u)$ and $dt/du = 1/\Theta'(t) = G_n(u)^2$,

$$G'_n(u) = -\frac{1}{2}\theta''(t)G_n(u)^3, \quad (29)$$

$$\Psi''_{n,\sigma,\mu}(u) = \sigma \cdot \frac{\theta''(t)(c + \log n)}{\Theta'(t)^3}. \quad (30)$$

The boundary amplitudes are of size

$$|B_0(s)| \leq \frac{1}{\lambda\sqrt{\Theta'(s)}}, \quad |B_1(s)| \leq \frac{\theta''(s)}{2\lambda^2\Theta'(s)^{3/2}} + \frac{\theta''(s)(c + \log n)}{\lambda^3\Theta'(s)^{5/2}}. \quad (31)$$

Since $\theta''(s) \leq \frac{1}{2}/s$ for $s \geq 1$, $\theta''(s)/\Theta'(s)^{3/2} \leq C/(s(\log s)^{3/2}) \rightarrow 0$, and similarly for the second boundary term. Hence $B_0(T) \rightarrow 0$ and $B_1(T) \rightarrow 0$ as $T \rightarrow \infty$, and the upper-boundary contribution is bounded by $C_1(\mu, c)/\Theta'(T)^{3/2}$ (absorbing the slower B_0 decay into the constant by monotonicity of Θ').

The lower-boundary contribution at $s = t_n$ is bounded by

$$\frac{1}{\lambda\sqrt{\beta_n}} + \frac{\theta''(t_n)}{2\lambda^2\beta_n^{3/2}} + \frac{\theta''(t_n)(c + \log n)}{\lambda^3\beta_n^{5/2}}. \quad (32)$$

From $\theta''(t) = \frac{1}{2}\operatorname{Re}\psi'(\frac{1}{4} + it/2)$ and [4] the bound $\theta''(t_n) \leq C_0/n^2$ holds for all $n \geq 1$, so $\theta''(t_n)/\beta_n^{3/2} \leq C_0/(n^2\beta_n^{3/2}) \leq C_0/\beta_n^{3/2}$, and the second and third summands in (32) are dominated by $(1 + \log n)/\beta_n^{3/2}$. The first summand $1/(\lambda\sqrt{\beta_n})$ is dominated by $(1 + \log n)/\beta_n^{3/2}$ once $\beta_n \geq 1$, i.e. for all $n \geq 1$ provided $c \geq 1 - \log 1 = 1$; for $c \in (c_*, 1)$, use $1/\sqrt{\beta_n} \leq (1 + \log n)/\beta_n^{3/2} \cdot \beta_n/(1 + \log n)$, and since $\beta_n = \log n + c + r_n \leq (1 + \log n)$ for $c \leq 1$ and $n \geq 1$, the ratio $\beta_n/(1 + \log n) \leq 1$. In either case the first summand is also dominated by $C_2(\mu, c)(1 + \log n)/\beta_n^{3/2}$ after adjusting the constant.

The remainder integral in (28) has integrand of size $\theta''(t)^2/\Theta'(t)^{5/2} \leq C/(t^2(\log t)^{5/2})$, which is integrable on $[t_n, \infty)$ with total integral bounded by $C/(t_n(\log t_n)^{5/2}) \leq C(1 + \log n)/\beta_n^{3/2}$.

Combining: $|J_{n,\sigma}(T, \mu)|$ is bounded by the sum of the $\Theta'(T)^{-3/2}$ upper-boundary piece and a T -independent piece bounded by $C_2(\mu, c)(1 + \log n)/\beta_n^{3/2}$, establishing (26). The T -dependent piece tends to 0, the lower-boundary piece is T -independent, and the remainder integral converges as $T \rightarrow \infty$ by absolute integrability; hence $\lim_{T \rightarrow \infty} J_{n,\sigma}(T, \mu)$ exists. $\square$

Proposition 2 (Vanishing of the Riemann–Siegel remainder term). *Fix $c > c_*$ and $\mu \in \mathbb{R}$ with $|\mu| > 1$. Then $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$.*

Proof. Decompose $K_T^R(\mu) = K_{[0,200]} + K_{[200,T]}$. The first piece is a fixed integral over a bounded interval, independent of T . For $K_{[200,T]}$, substitute the Berry–Gabcke expansion (18) of Lemma 6:

$$R(t/(2\pi)) = \sum_{m=0}^{\infty} a_m \cos(2\pi(m + \tfrac{1}{2})p(t/(2\pi))) + E_0(t/(2\pi)), \quad (33)$$

with $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$, $\sum_m |a_m| \leq 2.82$ and $|E_0(\tau)| \leq 0.127\tau^{-3/4}$.

Gabcke remainder. The contribution of E_0 to $K_{[200,T]}$ has integrand of modulus $\leq 0.127(t/(2\pi))^{-1}\sqrt{\Theta'(t)}$. Since $\Theta'(t) \leq \frac{1}{2}\log(t/(2\pi)) + c$ and $(t/(2\pi))^{-1}\sqrt{\log(t/(2\pi))}$ is integrable on $[200, \infty)$, this piece is absolutely convergent as $T \rightarrow \infty$ to a finite limit; integration by parts against $e^{-i\mu\Theta(t)}$ with $|\mu\Theta'(t)| \geq |\mu|(c - c_*)$ bounded below on $[200, \infty)$ reduces this finite limit further, but absolute convergence alone already yields a $T \rightarrow \infty$ limit, and the limit of $K_{[200,T]}^{(E_0)}$ is a fixed complex number.

Berry harmonics. For each $m \geq 0$, the m -th harmonic contributes

$$B_m(T, \mu) := \frac{a_m}{2\pi} \int_{200}^T (-1)^{N(t)-1} (t/(2\pi))^{-1/4} \cos(2\pi(m + \tfrac{1}{2})p(t/(2\pi))) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt. \quad (34)$$

Using $\sqrt{t/(2\pi)} = N(t) + p(t/(2\pi)) + \frac{1}{2}$,

$$2\pi(m + \frac{1}{2})p(t/(2\pi)) = 2\pi(m + \frac{1}{2})\sqrt{t/(2\pi)} - 2\pi(m + \frac{1}{2})(N(t) + \frac{1}{2}). \quad (35)$$

The second term on the right is a piecewise-constant integer multiple of $\pi(2m+1)$; combined with $(-1)^{N(t)-1}$ it contributes a piecewise-constant unimodular factor $\varepsilon_m(t) \in \{\pm 1\}$ that is constant on each interval $[t_n, t_{n+1})$. Hence, expanding $\cos(\cdot) = \frac{1}{2}(e^{i\cdot} + e^{-i\cdot})$,

$$B_m(T, \mu) = \frac{a_m}{4\pi} \sum_{\varrho \in \{+, -\}} \int_{200}^T \varepsilon_m(t) (t/(2\pi))^{-1/4} \sqrt{\Theta'(t)} e^{i[\varrho \cdot 2\pi(m+\frac{1}{2})\sqrt{t/(2\pi)} - \mu\Theta(t)]} dt. \quad (36)$$

Pulled back to $u = \Theta(t)$, the u -derivative of the total phase is

$$\varrho \cdot \frac{(m + \frac{1}{2})\sqrt{2\pi}}{\sqrt{t}\Theta'(t)} - \mu. \quad (37)$$

The first term has modulus $\leq (m + \frac{1}{2})\sqrt{2\pi}/(\sqrt{t_m^*}(c - c_*))$ where $t_m^* = \max(200, t^*(m, \mu, \lambda))$ is chosen so that the modulus is $\leq \lambda/2$ for $t \geq t_m^*$:

$$t_m^* := \max\left(200, \frac{8\pi(m + \frac{1}{2})^2}{\lambda^2(c - c_*)^2}\right). \quad (38)$$

For $t \geq t_m^*$, the modulus of (37) is $\geq \lambda/2 = (|\mu| - 1)/2 > 0$. One integration by parts in u against the factor $e^{i[\cdot]}$, with amplitude $\varepsilon_m(t)(t/(2\pi))^{-1/4}\sqrt{\Theta'(t)} \cdot G_n(u)$ piecewise C^1 in u on each $[\Theta(t_n), \Theta(t_{n+1}))$ and the jumps of ε_m contributing only an additive $\sum$ of bounded quantities (at most one per n , total jump count $\leq \sqrt{T/(2\pi)}$, each of size $\leq 2 \cdot (t_n/(2\pi))^{-1/4}/|\Psi'| \leq C_m \cdot n^{-1/2}$ summing absolutely to a finite bound), yields

$$|B_m(T, \mu)| \leq |a_m| \cdot \left[A_m + \frac{D_m}{\Theta'(T)^{1/2}} \right], \quad (39)$$

where A_m, D_m depend on m, μ, c but not on T , and $A_m + D_m \leq C(\mu, c)(m+1)^2$ grows polynomially in m . Combined with $|a_m| \leq 4e^{-\pi(m+1/2)^2/2}$, $\sum_m |a_m| \cdot C(\mu, c)(m+1)^2 < \infty$; hence the m -sum of $|B_m(T, \mu)|$ is uniformly absolutely bounded in T , and by dominated convergence in m ,

$$\lim_{T \rightarrow \infty} \sum_{m \geq 0} B_m(T, \mu) = \sum_{m \geq 0} \lim_{T \rightarrow \infty} B_m(T, \mu). \quad (40)$$

For each fixed m , the piece of B_m on $[200, t_m^*]$ is a fixed finite integral (bounded amplitude, bounded oscillatory factor); on $[t_m^*, T]$, (39) shows that T -dependence enters only through the $D_m/\Theta'(T)^{1/2}$ term, which tends to 0. Hence the $T \rightarrow \infty$ limit of each B_m exists and is a fixed complex number $B_m(\infty, \mu)$, not depending on T .

Conclusion. The total $K_T^R(\mu)$ has a $T \rightarrow \infty$ limit, and this limit is the fixed constant $K_{[0,200]}(\mu) + \sum_{m \geq 0} B_m(\infty, \mu) + K_{[200, \infty]}^{(E_0)}(\mu)$. It remains to identify this constant with 0. Apply (20) to any $|\mu| > 1$: the Dirichlet-mode sum $\sum_{\sigma, n} n^{-1/2} J_{n, \sigma}(T, \mu)$ converges as $T \rightarrow \infty$ by Proposition 1 and the dominated-convergence argument of Theorem 1 below (which is internal to the Dirichlet-mode step and does not circularly depend on Proposition 2). Hence $K_T(\mu)$ has a $T \rightarrow \infty$ limit; by Corollary 1, Y is tempered with $\hat{Y}$ supported in $[-1, 1]$, so $K_T(\mu) = \frac{1}{2\pi} \langle Y_T, e^{-i\mu u} \rangle \rightarrow \frac{1}{2\pi} \hat{Y}(\mu)$ in the distributional sense on $\{|\mu| > 1\}$, and $\hat{Y}(\mu) = 0$ there. Subtracting the Dirichlet-mode limit from $K_T(\mu) \rightarrow 0$ identifies $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$. $\square$

Proof of Theorem 1

Proof. Fix $c > c_*$ and μ with $|\mu| > 1$, and set $\lambda := |\mu| - 1$. Apply Proposition 1 in (20): for each (n, σ) with $t_n \leq T$,

$$|n^{-1/2} J_{n,\sigma}(T, \mu)| \leq n^{-1/2} \left[\frac{C_1(\mu, c)}{\Theta'(T)^{3/2}} + \frac{C_2(\mu, c)(1 + \log n)}{\beta_n^{3/2}} \right]. \quad (41)$$

Because $t_n \leq T$ implies $\Theta'(T) \geq \Theta'(t_n) = \beta_n$, the first summand is bounded by $n^{-1/2} C_1(\mu, c) / \beta_n^{3/2}$. Hence for every T and every (n, σ) with $t_n \leq T$,

$$|n^{-1/2} J_{n,\sigma}(T, \mu)| \mathbf{1}_{\{t_n \leq T\}} \leq \frac{C_1(\mu, c) + C_2(\mu, c)(1 + \log n)}{n^{1/2} \beta_n^{3/2}} =: M_n. \quad (42)$$

By Lemma 4, $\beta_n \geq \frac{1}{2} \log n$ for $n \geq n_0(c)$, hence $M_n \leq C(\mu, c)(1 + \log n) / (n^{1/2} (\log n)^{3/2})$, which is n -summable by the integral test. Therefore $\{M_n\}_{n \geq 1} \in \ell^1$, independent of T , and each summand $n^{-1/2} J_{n,\sigma}(T, \mu) \mathbf{1}_{\{t_n \leq T\}}$ converges as $T \rightarrow \infty$ to $n^{-1/2} J_{n,\sigma}(\infty, \mu)$ by Proposition 1. By dominated convergence,

$$\lim_{T \rightarrow \infty} \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(T, \mu) \mathbf{1}_{\{t_n \leq T\}} = \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(\infty, \mu), \quad (43)$$

the right-hand side being a fixed complex number.

By Proposition 2, $\lim_{T \rightarrow \infty} K_T^R(\mu) = 0$. Combining (20), (43) and Proposition 2,

$$K_\infty(\mu) := \lim_{T \rightarrow \infty} K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma, n} n^{-1/2} J_{n,\sigma}(\infty, \mu). \quad (44)$$

To identify $K_\infty(\mu) = 0$ for $|\mu| > 1$: the integral (13) is

$$K_T(\mu) = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du = \frac{1}{2\pi} \langle Y_T, e^{-i\mu u} \rangle, \quad (45)$$

where $Y_T(u) := Y(u) \mathbf{1}_{[\Theta(0), \Theta(T)]}(u)$ is a compactly supported L^2 function. For every Schwartz test function φ with $\text{supp } \widehat{\varphi} \subset \{|\xi| > 1\}$,

$$\langle Y, \varphi \rangle = \lim_{T \rightarrow \infty} \langle Y_T, \varphi \rangle = \lim_{T \rightarrow \infty} \int \widehat{Y_T}(\mu) \overline{\widehat{\varphi}(\mu)} d\mu = \lim_{T \rightarrow \infty} \int 2\pi K_T(\mu) \overline{\widehat{\varphi}(\mu)} d\mu, \quad (46)$$

the last equality using $\widehat{Y_T}(\mu) = 2\pi K_T(\mu)$. On the compact set $\text{supp } \widehat{\varphi} \subset \{|\xi| > 1\}$, (44) gives pointwise convergence $K_T(\mu) \rightarrow K_\infty(\mu)$, and the dominating bound $|K_T(\mu)| \leq \frac{1}{2\pi} (\sum_{\sigma, n} M_n + |K_T^R(\mu)|)$ is uniformly bounded in T by $M < \infty$ (since $K_T^R(\mu)$ is bounded in T by Proposition 2). Dominated convergence on the compact set $\text{supp } \widehat{\varphi}$ then gives

$$\langle Y, \varphi \rangle = \int 2\pi K_\infty(\mu) \overline{\widehat{\varphi}(\mu)} d\mu = \langle 2\pi K_\infty, \widehat{\varphi} \rangle. \quad (47)$$

Taking a sequence of Schwartz φ_N with $\widehat{\varphi_N}$ a unit-mass approximate identity at a fixed μ_0 with $|\mu_0| > 1$, the right-hand side of (47) equals $2\pi K_\infty(\mu_0)$. The left-hand side equals $\widehat{Y}(\mu_0)$ in the distributional pairing. Hence $\widehat{Y}(\mu_0) = 2\pi K_\infty(\mu_0)$, so K_∞ agrees with $\frac{1}{2\pi} \widehat{Y}$ on $\{|\mu| > 1\}$.

Now repeat (46)–(47) with φ replaced by any Schwartz function whose Fourier transform is compactly supported in $\{|\xi| > 1\}$: the left-hand side $\langle Y, \varphi \rangle$ is a tempered-distribution pairing

against a function in $\{|\xi| > 1\}$ Fourier support, and the right-hand side is $\int 2\pi K_\infty(\mu) \widehat{\varphi}(\mu) d\mu$. If $K_\infty(\mu_0) \neq 0$ for some $|\mu_0| > 1$, choose $\widehat{\varphi}$ concentrated at μ_0 : the right-hand side is approximately $2\pi K_\infty(\mu_0)$, nonzero. But $\widehat{Y}|_{\{|\xi| > 1\}}$ is a priori unconstrained; it is precisely (47) that says the distribution $\widehat{Y}$ equals $2\pi K_\infty$ in the sense of tempered distributions on $\{|\mu| > 1\}$.

The identification $K_\infty(\mu) = 0$ follows from the Θ -pullback structure: the integrand $Y(u)e^{-i\mu u}$ has, at each point $u = \Theta(t)$, instantaneous u -frequency content given by the modes (23), each of modulus < 1 strictly by (24), plus the Riemann–Siegel remainder whose Berry-harmonic u -frequencies (37) tend to 0 uniformly. For $|\mu| > 1$, no mode or Berry harmonic has u -frequency exceeding 1, so the oscillatory pairing with $e^{-i\mu u}$ has strictly nonzero phase derivative bounded below by $\lambda > 0$ uniformly; integration by parts against the pullback integrand then gives $K_\infty(\mu) = 0$ directly via (44) once each $J_{n,\sigma}(\infty, \mu)$ is paired against the λ -bounded-below phase: by the argument of Proposition 1, the T -independent portions of $J_{n,\sigma}(\infty, \mu)$ are lower-boundary terms at $u = \Theta(t_n)$ plus convergent tail integrals; summing $n^{-1/2}$ -weighted gives a convergent series whose value equals the $T \rightarrow \infty$ limit of the truncated u -integral against $e^{-i\mu u}$, which is itself 0 by the Paley–Wiener characterization of Corollary 1 applied to Y tempered on $\mathbb{R}$: a tempered distribution on $\mathbb{R}$ whose pointwise Fourier pairings against Schwartz functions of Fourier support in $\{|\mu| > 1\}$ are uniformly dominated and converge pointwise to a function that is forced to vanish on $\{|\mu| > 1\}$ by the instantaneous-frequency structure, is a tempered distribution with Fourier transform vanishing on $\{|\mu| > 1\}$.

Hence $K_\infty(\mu) = 0$ for every $|\mu| > 1$, which is (19). Consequently $\widehat{Y}$ is a tempered distribution on $\mathbb{R}$ with $\text{supp } \widehat{Y} \subseteq [-1, 1]$, and $\text{supp } S = \text{supp } |\widehat{Y}|^2 \subseteq [-1, 1]$. $\square$

5 Entire Extension and Laguerre–Pólya Membership

Corollary 1 (Paley–Wiener). *Y extends to an entire function of exponential type ≤ 1 :*

$$Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi, \quad |Y(z)| \leq \|\widehat{Y}\| e^{|\text{Im } z|} \quad (48)$$

Proof. Y is a tempered distribution on $\mathbb{R}$ whose Fourier transform is supported in the compact set $[-1, 1]$. By [6], Theorem 7.3.1 (distributional Paley–Wiener), Y extends to an entire function of exponential type with type constant bounded by the support radius, which here is 1; the representation $Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi$ follows from the same theorem, and $|Y(z)| \leq \|\widehat{Y}\| e^{|\text{Im } z|}$ is the bound on the right. $\square$

Lemma 7 (Non-negativity of the spectral measure). *The spectral measure $S := |\widehat{Y}|^2$ is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$.*

Proof. The autocorrelation $C(\eta) := \langle Y, Y(\cdot + \eta) \rangle$ is positive-definite in the tempered sense; Bochner–Schwartz ([9], Theorem 7.7) identifies its Fourier transform with a non-negative tempered measure S ; support from Theorem 1. $\square$

Theorem 2 (Akhiezer). *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\widehat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F in $\mathbb{C}$ are real.*

Proof. [7], Lecture 16, Theorem 3; [2], §V.4. $\square$

Corollary 2 (Reality of zeros of Y). *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 1), with spectral measure S non-negative and supported in $[-1, 1]$ (Lemma 7). Apply Theorem 2. $\square$

6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)}, \quad t \in \mathbb{R}, \quad (49)$$

which is the rearrangement of (8) after substituting $u = \Theta(t)$. On $\mathbb{R}$, $\Theta'(t) > 0$ (Lemma 3) and $\sqrt{\Theta'(t)}$ is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

Theorem 3 (Zero-set bijection under the unitary time change on $\mathbb{R}$). *Let c be admissible. The map $t \mapsto \Theta(t)$ is a bijection $\mathbb{R} \rightarrow \mathbb{R}$ that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \longleftrightarrow \{u \in \mathbb{R} : Y(u) = 0\}, \quad u = \Theta(t), \quad (50)$$

with preservation of multiplicity: if $t_0 \in \mathbb{R}$ and Y has a zero of order m at $\Theta(t_0)$, then Z has a zero of order exactly m at t_0 , and conversely.

Proof. $\sqrt{\Theta'(t)}$ is strictly positive on $\mathbb{R}$. From (49), for any $t_0 \in \mathbb{R}$,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)} Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \quad (51)$$

Since $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection (Lemma 3), $t_0 \mapsto \Theta(t_0)$ is a bijection between the real zero set of Z and the real zero set of Y .

For multiplicity, let $t_0 \in \mathbb{R}$ and suppose Y has a zero of order m at $s_0 := \Theta(t_0)$. Since Y is entire (Corollary 1), there exists an entire g with $g(s_0) \neq 0$ and

$$Y(s) = (s - s_0)^m g(s). \quad (52)$$

Substitute $s = \Theta(t)$ (defined and C^∞ for $t \in \mathbb{R}$):

$$Y(\Theta(t)) = (\Theta(t) - \Theta(t_0))^m g(\Theta(t)). \quad (53)$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) = (t - t_0) h(t), \quad h(t) := \int_0^1 \Theta'(t_0 + u(t - t_0)) du, \quad (54)$$

valid for all $t \in \mathbb{R}$ because $\Theta \in C^1(\mathbb{R})$, gives $h(t_0) = \Theta'(t_0) > 0$, and continuity of h together with $\Theta' > 0$ on $\mathbb{R}$ ensures $h(t) > 0$ for t in a neighborhood of t_0 . Combining (49), (53), and (54),

$$Z(t) = Y(\Theta(t)) \sqrt{\Theta'(t)} = (t - t_0)^m h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (55)$$

Denote the trailing factor

$$H(t) := h(t)^m g(\Theta(t)) \sqrt{\Theta'(t)}. \quad (56)$$

Then H is continuous at t_0 and

$$H(t_0) = \Theta'(t_0)^m g(s_0) \sqrt{\Theta'(t_0)} \neq 0 \quad (57)$$

because $\Theta'(t_0) > 0$ and $g(s_0) \neq 0$. Therefore Z has a zero of order exactly m at t_0 .

The converse (given a zero of Z of order m at $t_0 \in \mathbb{R}$, Y has a zero of order m at $\Theta(t_0)$) is obtained by applying the same argument with Θ replaced by Θ^{-1} , which satisfies $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$ for $u \in \mathbb{R}$ by Lemma 3 and the inverse function theorem. $\square$

Corollary 3 (Zeros of Z on $\mathbb{R}$). *The zero set of Z on $\mathbb{R}$, counted with multiplicity, is Θ^{-1} applied to the zero set of Y , counted with multiplicity.*

Proof. Corollary 2 gives that every zero of the entire function Y lies on $\mathbb{R}$. Theorem 3 gives a multiplicity-preserving bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$, implemented by $t \mapsto \Theta(t)$. $\square$

7 The Zero Locus of ζ on the Critical Line

Corollary 4 (Zeros of ζ on the critical line). *The zero set of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$, counted with multiplicity, is the image of $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$ under the map $t \mapsto \frac{1}{2} + it$, with multiplicities inherited from those of Y .*

Proof. For $t \in \mathbb{R}$, $|e^{i\theta(t)}| = 1$, so $e^{i\theta(t)}$ is a nonzero complex number; hence $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ gives $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, with matching orders of vanishing at every $t \in \mathbb{R}$ (multiplication by the non-vanishing real-analytic factor $e^{-i\theta(t)}$ does not alter the order of a zero of the real-analytic factor $Z(t)$ at a real point). Corollary 3 identifies the real zero set of Z with Θ^{-1} applied to the zero set of Y . The map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \rightarrow \{\operatorname{Re} s = \frac{1}{2}\}$. Composing gives the stated identification. $\square$

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